



MECH MAINT- TURBINE Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2024-25/LMI_HEAT/1036547

Date:
12/03/2025(19:08:10)

Subject:Approval for LMI on ‘Care and Maintenance of Turbine Plant Feed System’

CC-OS has revised the OGN with ref.no: COS-ISO-00-OGN/OPS/MECH/031 Rev. No.: 3 Jan 2025. The OGN supersedes OGN/OPS/MECH/031 Rev. No. 2 June 2022 – ‘Care and Maintenance of Turbine Plant Feed System’.

So, the LMI on ‘Care and Maintenance of Turbine Plant Feed System’ with Ref no: LMI/OGN/OPS/MECH/031 Rev. - 04 is due for revision and necessary changes are done in the LMI.

LMI is hereby put up for approval of Competent Authority.

Changes incorporated in this revision (Rev: 5) of LMI, have been mentioned as below

Sl No	LMI/OGN/OPS/MECH/031 REV.-04 Jul 2024	LMI/OGN/OPS/MECH/031 REV.-05 Mar 2025
1.	4.2.5 (v) Eddy current testing of tube shall be carried out after completing 1.5 Lakhs running hours or if repeated failure of tube leakage	Eddy current testing of heater tube shall be carried out after completing: - - For LP Heaters: 1.5 Lakhs running hours or if repeated failure of tube leakage. - For HP heaters: 10 years or if repeated failure of tube leakage. (As per AGM(COS-ST) IOM COS/ST/GEN dt. 07.01.2008)

		After first eddy current testing of heater tubes, subsequent testing of tubes to be carried out as per recommendations or in every COH whichever is earlier.
2.	4.4.2 (vi) -	Heaters tube leakage test by pressurizing the shell side with air to be carried out in every OH
3.	Annexure B6 -	Checklist for Care and Maintenance of Turbine Feed System

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LMI is in line with the OGN.

May please be approved.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**CARE AND MAINTENANCE OF TURBINE PLANT
FEED SYSTEM**

LMI/OGN/OPS/MECH/031

MARCH-2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	CARE AND MAINTENANCE OF TURBINE PLANT FEED SYSTEM
2	LMI No.	LMI/OGN/OPS/MECH/031 REV.-05
3	Reference Document	COS-ISO-00-OGN/OPS/MECH/031 Rev. No.: 3 Jan 2025
4	Superseded Document	LMI/OGN/OPS/MECH/031 REV.-04
5	Prepared By	Name: Sahil Sankalp, Manager (TMD)
6	Checked By	Name: Nilanjan Mandal, AGM (TMD)
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8	Approved By	Name: Sameer Sharma H O P (Simhadri)

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TITLE	Care and Maintenance of Turbine Plant Feed System
LMI No.	LMI/OGN/OPS/MECH/031

1.0 INTRODUCTION

The feed system is defined as the complete feed water heating train, including all stages of feed heating, associated condensate and feed pipework and valves, bled steam pipework and valves, drain pipework, valves and receivers but excluding the main pumping units and their respective drives (e.g. condensate extraction pumps, drip pumps and boiler feed pumps).

For the purpose of this document, the feed heaters currently in service will be sub-divided into three groups:

- i. Low pressure (LP) heaters, surface type
- ii. Deaerators
- iii. High pressure (HP) heaters, surface type

2.0 SUPERSEDED DOCUMENTS

LMI/OGN/OPS/MECH/031 REV.-04 Jul 2024

3.0 DETAILS OF REVISION

Following changes have been incorporated in the current revision –

SI No	LMI/OGN/OPS/MECH/031 REV.-04 Jul 2024	LMI/OGN/OPS/MECH/031 REV.-05 Mar 2025
1.	4.2.5 (v) Eddy current testing of tube shall be carried out after completing 1.5 Lakhs running hours or if repeated failure of tube leakage	Eddy current testing of heater tube shall be carried out after completing: - - For LP Heaters: 1.5 Lakhs running hours or if repeated failure of tube leakage. - For HP heaters: 10 years or if repeated failure of tube leakage. (As per AGM(COS-ST) IOM COS/ST/GEN dt. 07.01.2008) After first eddy current testing of heater tubes, subsequent testing of tubes to be carried out as per recommendations or in every COH whichever is earlier.
2.	4.4.2 (vi) -	Heaters tube leakage test by pressurizing the shell side with air to be carried out in every OH
3.	Annexure B6 -	Checklist for Care and Maintenance of Turbine Feed System

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4.0 RECORDS OF FEED SYSTEM STATUS:

4.1 Objective

The objective is to provide a database for future engineering upgrades.

4.2 Information Required

The comprehensive list presented in sub-section 4.2.1 to 4.2.5 inclusive below identifies all the data that may be required to assess the status of a feed system. A database should be selected from this list having due regard to the level of information available from manufacturers for each individual system.

4.2.1 Original Design Intent (Data shall be maintained by TMD)

- i) Heater types
- ii) Diagrammatic arrangement of feed system
- iii) Valve schedule
- iv) General arrangements of each heater
- v) Detail arrangements of:
 - a) Heater shells (including supports and bolting)
 - b) Tube nests (including baffles, shrouds, etc.)
 - c) Tube plates/heaters including tube fixings
 - d) Water box/division plate
 - e) De-aerating head/tray work and/or spray nozzle details
 - f) Bled steam pipework
 - g) Condensate/feed pipework
 - h) Drainage pipework (including orifices if fitted)
 - i) Drains receivers
 - j) Pressure relief valve capacities and settings
 - k) Quality limits for pipework bends
- vi) Operating conditions for each stage (including tapping points and conditions for bled steam).
- vii) Recommended rates of temperature change across heavy sections.

4.2.2 Modification to Original Design (Data Shall be maintained by TMD and C&I)

Include details, with full supporting drawings, of all major modifications to the original design e.g. pipework/valve modifications, additional valves, basic design changes to the heaters themselves, methods of de-superheating, drains cooling, routing of drainage etc.

Where modifications to the Control and Protection System involve modifications to hardware these should also be recorded.

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4.2.3 Manufacturing Information (Data Shall be maintained by TMD)

Material specification for:

- i) Heater shells/tubeplates/headers
- ii) Tubes
- iii) Condensate / feed pipework
- iv) Bled steam pipework
- v) Valves

4.2.4 Operational History (Data Shall be maintained by EEMG)

- i) Date of installation of original system.
- ii) Date of installation of existing system where it differs from the above.
- iii) Operating hours of last inspection.
- iv) Numbers of hot, warm and cold unit starts at last inspection, with best judgement of feed train starts, as defined in Annexure A.
- v) Typical maximum temperature gradient across heavy sections under past operating regime in case measurements are available at stations.
- vi) A summary of problem areas and failures experienced to date.

4.2.5 Inspection and Maintenance (Data Shall be maintained by TMD)

1. Details of past inspections of
 - a) Heater internals
 - b) Feed water isolating valves,
 - c) Drip line valves,
 - d) Extraction valves,
 - e) Safety relief valves etc.
carried out during unit overhaul or any available opportunity.
2. Summary of findings in each case including:
 - a) Component/area inspected
 - b) Method of inspection
 - c) Results
 - d) Action taken
 - e) Details of Hydro test carried out (shell side) & findings.
3. Components replaced with date
4. Date of next planned inspection
5. Current estimate of remnant life after carrying out of Eddy current testing of tubes.
Eddy current testing of heater tube shall be carried out after completing: -
 - a) For LP Heaters: 1.5 Lakhs running hours or if repeated failure of tube leakage.

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b) For HP heaters: 10 years or if repeated failure of tube leakage. (As per AGM(COS-ST) IOM COS/ST/GEN dt. 07.01.2008)

After first eddy current testing of heater tubes, subsequent testing of tubes to be carried out as per recommendations or in every COH whichever is earlier.

6. Thickness survey of drip line pipes, ends, reducer etc. during every unit overhaul and replacement of the same if thickness reduction found below 10%.
7. Date of past cleaning operations including:
 - a) Area cleaned (with details of isolation required)
 - b) Methods employed with details of chemicals used, if any
 - c) Results
 - d) Details of any before-and-after performance monitoring
8. Date of next planned cleaning operation
9. For surface-type heaters only:
 - a) Date of tube failures plus location including height above/from tube-plate face if measured.
 - b) Details of tube plugged plus method used
 - c) Updated diagram of plugging situation

4.3. Engineering aspects of plant operation

4.3.1. Objectives

To define the instrumentation required:

- i) To provide operating limits that will minimize component deterioration, and
- ii) To provide essential protection for other areas of the plant from fault conditions arising in the feed system.

4.3.2. The influence of operating conditions on component life

As a complex, integrated system, the feed system components can be subjected to two distinct types of life-limiting mechanisms. These can be classed as:

- i) Direct mechanisms, i.e. those caused directly on a component by off- design excursions out with the feed system.
- ii) Indirect or knock-on mechanisms where a component is subject to deterioration due to a fault condition in another part of the feed system.

These two types of mechanisms will be dealt with separately.

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4.3.2.1 Direct Acting Mechanisms

The major direct-acting mechanism affecting feed heater life is thermal fatigue.

The rate of change of surface temperature on the heavy sections, particularly of HP surface-type heaters can give rise to excessive temperature gradients across the section. This coupled with stress concentrating features such as small radial curves or coarse machining marks can lead to high strain, low cycle fatigue cracking.

In extreme cases, rapid heating or cooling transients, coupled with existing cracks or defects, can lead to fast brittle fracture.

4.3.2.2 Indirect Mechanisms (Data Shall be maintained by TMD)

Some examples of the indirect mechanisms involved in limiting component life in the feed system are:

- i) Tube fouling
- ii) Tube failure
- iii) Failure of internal shrouds
- iv) Failure of adequate drains control
- v) Over pressurization
- vi) Failure of shell joints
- vii) Failure of water box division(partition) plate integrity/sealing

Fouling on either surface will reduce the Heat Transfer Coefficient (HTC) of the nest, thus reducing the feed outlet temperature and the degree of drains cooling (if any). The effect of the former on subsequent heaters is to increase their required bled steam flow, with the possibility of vibration damage to tubing and steam inlet geometry.

The effect of the latter on drains systems is to lead to flashing and consequential cavitation erosion in drain lines, orifices, valves and receivers.

Tube failure is the most common cause of heater life limitation, in that a finite element of heat transfer surface is removed with each plugged tube. Causes of failure vary from erosion due to high local velocities of condensate droplets to vibration-induced wear either on fixed baffle plates or, in extreme cases, by tube 'clashing': at anti-nodal points.

The consequences of tube failure are to limit the life of the heater concerned and to apply a greater load to subsequent heaters.

The failure of internal shrouds such as those which delineate de-superheating or drains cooling sections of HP Heaters usually results from high strain, low cycle fatigue induced by bad design coupled with differential expansion rates.

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Shroud failure can lead to tube damage by the introduction of condensate droplets into desuperheater zones, and to complete loss of drains cooling due to the introduction of saturated condensate and/or steam into the drainage leaving the shell. This in turn will have serious erosion effects on the whole of the drainage management system.

Adequate drainage level control is essential on all feed systems. Too high a level will lead to an excessive number of heater trips with consequential thermal shock transients; too low a level will lead to two-phase flow in the drainage system, with consequential damage already detailed.

The over pressurization can occur in deaerators when the HP heater drains are led directly to the deaerator via orifice plates. Failure, or even major wear, of these orifices could lead to the deaerator being over pressurized at a rate with which the relief valves are unable to cope. This is potentially dangerous.

Failure of the integrity of the water box division plate (partition plate), or its seal, leads to involuntary bypassing of the tube nest and consequent loss of feed outlet temperature.

4.3.3. Essential Protection for Other Plant from Faults Originating in the Feed System

The greatest risk to other plant from feed system faults is the possibility of water, either feed or condensed bled steam, being inducted into the turbine (either a main turbine or a Boiler Feed Pump Turbine (BFPT)).

The situation can arise from either a blocked heater drains system or a multiple tube failure in a surface type heater. The ingress of water into turbine and the on-load testing requirements detailed therein, particularly with regard to the duplicate float switches, should be rigidly adhered to, for the on-load testing of bled steam non- return valves.

4.3.4. Required Monitoring Instrumentation (Data Shall be maintained by EEMG /OPERATION)

Many of the life-limiting mechanisms can only be detected, on-load, by their secondary effects, such as a lower feed temperature rise across a given heater, or a loss of sub-cooling in the drainage flow.

Provision should be made for the following parameters to be monitored in service using suitable pressure tapping and thermocouples, on all feed heaters

- i) Feed water inlet temperature
- ii) Feed water outlet temperature
- iii) Feed water flow (per train if available otherwise total feed flow)
- iv) Bled steam temperature at heater inlet
- v) Heater shell pressure
- vi) Drains outlet temperature

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All the above should be available to the operator and recorded by the Station on a monthly basis for future analysis/audit purposes.

In addition, the condensate level in all heaters should be clearly indicated local to the heater and, desirably, at the operator's control desk.

4.3.5. Interim Arrangements (Data Shall be maintained by EEMG/OPERATION)

In the absence of automatic data logging equipment, the following manual records should be kept for all available monitoring positions monthly:

- i) Feed water inlet temperature
- ii) Feed water outlet temperature
- iii) Feed water flow
- iv) Bled steam temperature
- v) Heater shell pressure
- vi) Drains outlet temperature

4.3.6. Operating Limits (Responsibility: Operation Department)

The rate of returning the feed system, or any part of the feed system to service on the feed side should be governed by the rate of temperature change across the heavy sections.

Operating instructions should be drawn up for each station, based on data on heater metal temperatures versus feed inlet temperatures, gathered from the instrumentation.

The operating instructions should be defined in terms of length of preceding system outage and available feed water inlet temperature.

The rate of application of bled steam will be dictated by the capacity of the drains control system to maintain control of heater levels within the normal operating range.

4.4. Plant Maintenance and Modification (Responsibility: TMD)

4.4.1. Objectives

To identify the content and timing of planned maintenance, to outline a procedure for formulating cleaning/inspecting/repair/refurbishment schemes; to identify the considerations regarding the introduction of modifications.

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4.4.2. Planned Maintenance Activities

These should be based on the results of previous inspections and on problems identified on similar type of heaters.

When heaters and their associated pipework are opened up on the waterside, particular attention should be paid to:

- i) Integrity of feed inlet, outlet and bypass isolating valves (valve passing & Rectification).
- ii) The condition of pipework, and adequate support systems.
- iii) NDT inspection of the areas of the water box. Circumferential skirt/water box weld roof, division plate support bracket welds, internal areas adjacent to branch pipes, access door recess, water box to pipe welds, external junction of water box and branch pipes and tubeplate ligaments and tube plate.
- iv) Tube internal cleanliness / fouling. The need for chemical and/or mechanical cleaning of tubes shall be dictated by:
 - a) The degree and type of fouling discovered and,
 - b) Whether or not a tube-wall thinning NDT inspection is required
- v) Any particular emerging pattern of tube failures, the degree and rate of tube failure may indicate the need for tube-wall thinning survey using appropriate technique.
- vi) Heaters tube leakage test by pressurizing the shell side with air to be carried out in every OH.

On the deaerators particular attention should be paid to the inspection of structural welds together with selected branch and attachment welds. The results of these inspections should be recorded in the feed system status records (Annexure-B1 to B5).

4.4.3. Preventive Maintenance

Where a particular area of a nest has been identified as being prone to on- load tube failure, a tube wall thinning inspection should be carried out on this and other similar heaters.

Depending on the rate of tube failures and on the original tube wall thickness, a level 'allowable wear' will be established, with all tubes detected with greater wear being sacrificially plugged.

4.4.4. Modification

When a new heater is required, the following options shall be considered:

- i) Re-tube of the existing tubeplate / header to the original nest design.
- ii) Re-tube of the existing tubeplate / header to a modified nest design, particularly as to de-superheating and drains cooling arrangements.

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Note: Both of the above options are dependent on the tubeplate / header integrity being established by comprehensive NDT.

- iii) Re-tube of a new tubeplate / header but using the existing shell.
- iv) Re-tube of a new tubeplate / header but re-using the water-box.
- v) Replacement of a complete heater to the original design.
- vi) Replacement of a complete heater to a new design to improve life / integrity.

The final selection should be based upon economic considerations; timescale and limits imposed by the existing pipework and supporting steelwork and foundation structures.

4.5. Life assessment (Responsibility: TMD)

4.5.1. Objectives

To define a methodology for determining the degree of component degradation and identify the action required to assure continued safe, efficient operation to a predictable limiting life. RLA (Residual life assessment) may be carried out after completing 1.5 Lakhs running hours.

4.5.2. Cumulative Tube failure

While not affecting plant safety, the cumulative effect of tube failures on heater efficiency (as indicated by terminal temperature difference) is the commonest cause of life limitation. Hence the justification for plant replacement in this case will be made on economic grounds.

An assessment should be made, for each stage of feed heating, of a datum level of percentage surface loss above which a plant replacement scheme might become viable. This assessment should also take into account both current and predicted rates of tube failures, and probable lead-time required to obtain replacement plant.

The results of this assessment for each heater should be entered in the feed system status records.

Once any heater has reached its datum level, a specific assessment of the financial case for its replacement should be carried out. Should this case fail to meet current criteria, the exercise should be repeated at least annually, until such time as a case can be made.

4.5.3. Crack Growth and Thermal Fatigue

Typical temperature profiles for hot, warm and cold starts from the plant status records should be used with current high strain fatigue endurance data to predict the number of each type of start cycle required for critical crack depth. This should be done on one heater

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per station.

The assessed life should then be calculated for each affected heater using the best data available on the actual number of cycles experienced by the plant.

Current advice that cracks should not be ground out, other than short sections for the purpose of crack sizing if other sizing methods are considered inadequate. The grinding should be continued to a depth of 2mm below the crack tips to remove the most damaged material and every effort should be made to provide smoothly radiused profiles.

A datum level NDT inspection and assessment of current crack depth should be carried out on each heater water box / header as soon as is practicable, and the results noted in the feed system status records. Thereafter similar inspections and assessments should be carried out as necessary.

Experience has shown that the initiation of thermal fatigue cracking is not necessarily indicative of imminent component failure.

The assessment of the future life of the heavy section should therefore be based upon the appraisal and monitoring of any cracks found in the outage inspection, in comparison with the previously noted datum figures. The defect appraisal should be carried out before the unit is returned to service to determine the need for any immediate action, and the interval before re- inspection.

The defect appraisal assessment should take into account the past and anticipated future operational patterns, manufacturing and repair details, and inherent design integrity. The assessment should be based upon past experience, stress analysis, fractographic and metallurgical studies as appropriate. The results should be included in the feed system status records.

4.5.4. Inspection Procedures

The tube-wall thinning inspection and the crack depth measurements should be carried out to defined procedures. The procedures should cover the NDT techniques to be used, the measurements required and the method of recording.

The results of all inspections should be included in the feed system status records.

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5.0 RECORDS TO BE MAINTAINED

Sl No.	Record Description	Maintained By	Frequency	Remarks
1	Records must be kept for all inspections and should contain the following data: • Date of inspection • Details of inspection carried out • Date of testing & details of testing • Exception of the inspection. • Detailed of repair carried out.	Responsibility of carrying out different activities shall be as per given above. Joint protocol shall be made by TMD along with FQA for all weld repairs & NDT inspections. Joint protocols shall be made with Operation department after attending all defects & thorough cleaning before final box up.	As mentioned, Above.	All defects noticed during above inspection and testing shall be rectified during overhaul / available opportunity.

6.0 REVIEW

This LMI document shall be reviewed by the HOD-TMD on a 2 yearly basis or as necessary and the same shall be approved by HOP.

7.0 DISTRIBUTION OF LMI

Soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG Department.

8.0 LMI CHECKLIST

Checklist for Care and Maintenance of Turbine Feed System

9.0 LIST OF ANNEXURES

Annexure A: DEFINITION OF START CATEGORIES

Annexure B1: FORMAT OF TURBINE FEED SYSTEM STATUS RECORDS

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Annexure B2: FEED SYSTEM STATUS RECORD

Annexure B3: OPERATIONAL HISTORY

Annexure B4: INSPECTION AND MAINTENANCE

Annexure B5: RUNNING RECORDS

Annexure B6: Checklist for Care and Maintenance of Turbine Feed System

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ANNEXURE A**DEFINITION OF START CATEGORIES**

(Data shall be maintained by EEMG)

The categories defined below, represent best current practice and shall be adopted for all future plant. However, where another categorization is already established a change is not required but the categorization employed must be defined in the feed system status record.

- i) Cold start, after a shutdown period exceeding 72 hrs. (metal temperatures below 40% of their full load values in °C).
- ii) Warm start, after a shutdown period of between 10 hrs. and 72 hrs. (metal temperatures between 40% and 80% of their full load values in °C).
- iii) Hot start, after a shutdown period of less than 10h (metal temperatures above 80% of their full load values in °C).
- iv) Very hot restart, within 1hr. after a unit trip or after feed heater bank trip (metal temperatures at or near their full load values).

Note: If these are not recorded in a particular station then records (i) to (iii) are sufficient.

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ANNEXURE B1: FORMAT OF TURBINE FEED SYSTEM STATUS RECORDS

A single unified format for the feed system status record is not appropriate in view of the wide variation in circumstances that will exist.

The following format is presented as an adequate example, for an imaginary case where the importance of the plant involved has justified a full implementation of this code of practice, but where no major hazard is recognized and where the life limitation is already identified with adequate engineering and budgetary provision in hand.

It will be seen that the suggested format does not call for all relevant documentation to be brought together. Only documents necessary for the ongoing operation and engineering matters are appended to the document.

In other circumstances such as where one or more heaters are seen as being at risk and in need of investigation it will be necessary to bring together all the information for comprehensive investigation. The other extreme will be where a permanent dispensation regarding implementation has been agreed.

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ANNEXURE B2: FEED SYSTEM STATUS RECORD**(Data shall be maintained by Operation)**

STEAM SOURCE	HEATER	DESIGN PARTICULARS
Ext 1	LPH-1	
Ext 2	LPH-2	
Ext 3	LPH-3	
Ext 4	TO BFPT / D/A	
Ext 5	HPH-5 A/B	
Ext 6	HPH-6 A/B	

MANUFACTURER:

HEATER TYPES:

DRAIN COOLERS:

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ANNEXURE B3: OPERATIONAL HISTORY**(Data shall be maintained by EEMG & TMD)**

- i) Units inspected and dates
- ii) Operating hours at last inspection
- iii) Number of hot, warm and cold starts to date
- iv) Maximum temperature gradients across heavy sections under past operating regimes in case measurements are available at stations.
- v) Feed Heaters Performance Test Records. Test frequency is monthly
- vi) Quarterly with online instrument and yearly with offline instruments and is carried by Station EEMG.

PROBLEM AREAS

PLANT	PROBLEM
All HP heaters	Water box circumferential cracking HP heater drains
	Pipe erosion and failure of NRV

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ANNEXURE B4: INSPECTION AND MAINTENANCE**(Data shall be maintained by TMD)**DATE AND UNIT
INSPECTED

COMPONENT/ AREA

FILE/ REPORT NOS.
ALTERNATIVE SOURCES**RECORD OF DEFECTS**

UNIT	EQUIPMENT	PARTING PLANE INSPECTION	LAST SHELL SAFETY VALVE SERVICING	TUBE LEAKAGE/ TUBE PLUGGING IF ANY	VALVE PASSING (INLET/ OUTLET/ BYPASS VALVE) IF ANY

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ANNEXURE B5: RUNNING RECORDS**(Data shall be maintained by Operation)**

UNIT LOAD MW FEED FLOW Kg/s

Heater

FEED IN (°C)

FEED OUT (°C)

EXTN STEAM IN (°C)

EXTN PRESSURE (°C)

DRAINS (°C)

SHELL PRESSURE (KSC)

Heater

DC

Deaerator

FEED IN (°C)

FEED OUT (°C)

EXTN STEAM IN (°C)

EXTN PRESSURE (°C)

DRAINS (°C)

SHELL PRESSURE (KSC)

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ANNEXURE B6: Checklist for Care and Maintenance of Turbine Feed System

S No	Points for Observation	Responsibility	Checked (Yes/No)	Remarks
1.	Check following Manufacturing information maintained: • Material of Construction for Shell, Tubeplates, tubes, piping, valves, etc. • Hanger and support details • Known commissioning defects	TMD		
2.	Check following original design data is maintained: • Details of heater type, general arrangement, valve schedule, layout, etc. • Details of deaerator head, tray, spray nozzles, etc. • Piping details • Vent orifice size • Pressure relief valve settings	TMD		
3.	Check information of any modification to original design carried out is maintained	TMD/ C&I		
4.	Check details of operational limits are maintained: • Extraction Steam pressure and temperature • Drip Levels in heat exchangers and alarm/trip values. • Terminal parameters for heat exchangers for Feed water and extraction/drip • Gain in Feedwater across heat-exchanger • Essential protection to safeguard equipment	Operation		
5.	Check required monitoring instrumentation is maintained such as: • Feed water Inlet/Outlet Temperatures • Feed water flow (per train or total) • Extraction Steam Temperature • Heater shell pressure • Drain Outlet temperature	EEMG/ Opn		
6.	Check Operational history is maintained: • Number of Hot, warm and cold unit starts since last inspection	EEMG		

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	- Operating hours since last inspection - Summary of problem areas and failures experienced - Performance test reports to indicate shortfall in efficiency and likely reasons - Losses on account of short circuiting, improper insulation, etc.			
7.	Check inspection and maintenance history is maintained: - Past inspection details - Integrity of feed inlet, outlet and bypass isolating valves (valve passing & Rectification). - The condition of pipework, and adequate support systems. - Details of NDT inspections carried out. - Thickness survey of pipeline, DPT of weld joints, etc. - Details of tube leakage test & tubes plugged with graphical representation. - Details of Eddy current testing results & %of thinning observed.	TMD		
8.	Check Life assessment to be carried out after 1.5 Lakh running hours and inspection to assess crack formation and thermal fatigue	TMD		
9.	Eddy current testing of HPH & LPH tubes shall be carried out after completing 10 years and 1.5 Lakhs running hours respectively or if repeated failure of tube leakage. After first testing, eddy current testing of tubes to be carried out in every COH.	TMD		
10.	Thickness survey of drip line pipes, ends, reducer etc. during every unit overhaul and replacement of the same if thickness reduction found below 10%	TMD		
11.	Following inspections are to be carried out during every unit overhaul: - Inspection of Heater internals, Feed water side water box parting plane inspection to find out any erosion. - Complete replacement of gland packing set of main feed water valve including IBV during every unit overhaul. Gland packing shall be of suitable class /rating, pure	TMD TMD		

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	graphite endless ring with profile cross section as recommended valve OEMs • NDT examination of weld joints of feed water valve IBVs. Thickness mapping of piping of IBVs. • Valve operation check for freeness and valve open-close limit / torque setting • Servicing of actuator gear box for all feed water valves. • Servicing all Feed water control valves including replacement of gaskets /soft components. • Checking of valve isolation shall be done before overhaul, in case of any passing / defects complete overhauling of valve may be taken up during unit overhaul. • Inspection of drip line control valves, Extraction valves, Safety relief valves etc.	TMD Opn/ EMD TMD TMD Opn/ TMD TMD		
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